Summary in Graph

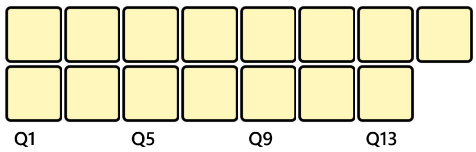
Exam Summary (GO Classes DA Test Series 2026 | Artificial Intelligence | Topic Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (53) out of 525 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider a search problem $(S, A, Succ, s_0, G)$, where all actions have cost 1. S is the set of states, $A(s)$ is the set of legal actions from a state s, $Succ(s, a)$ is the state reached after taking action a in state s, s_0 is the start state, and $G(s)$ is true if and only if s is a goal state.
Suppose we have a search problem where we know that the solution cost is exactly k, but we do not know the

actual solution. The search problems has $|S|$ states and a branching factor of b.
Unfortunately, we get an out of memory error when we try to use breadth first search.
Which of the following algorithms is the best one to use instead?

A. Depth First Search
B. Depth First Search limited to depth k.
C. Iterative Deepening
D. Uniform Cost Search

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Which of the following statements is/are TRUE?

A. For a search problem, the path returned by uniform cost search may change if we add a positive constant C to every step cost.
B. Uniform-cost search will never expand more nodes than A^*-search
C. Depth-first search will always expand more nodes than breadth-first search.
D. A* search will always expand fewer search nodes than uniform cost search

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following algorithms is/are guaranteed to give an optimal solution (given no negative edges)?

A. Greedy Best First Search
B. A^* with zero heuristic
C. A^* with consistent heuristic
D. Depth First Search

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

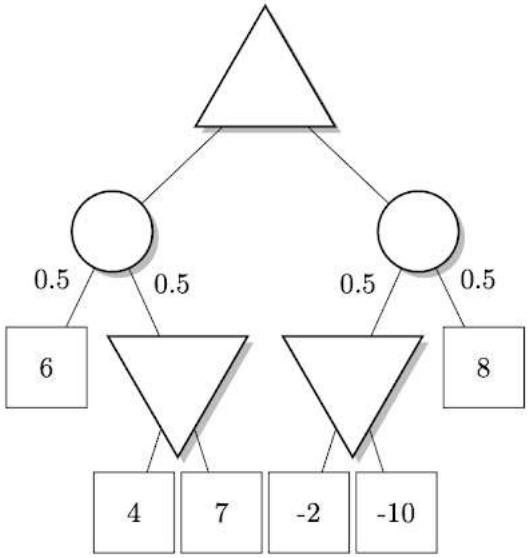
For a game tree where each node has b children and a d-ply lookahead is performed, what is the number of leaf nodes to be examined?

A. $b * d$
B. b^d
C. d^b
D. $b + d$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the two-player zero-sum game with elements of chance represented by the following tree. Triangles pointing up are max nodes, triangles pointing down are min nodes, circles are chance nodes with the probabilities of reaching the next node reported on the outgoing edges, and squares are terminal nodes with the corresponding value of utility function for max player.



What will be the expectiminimax value of the root node?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

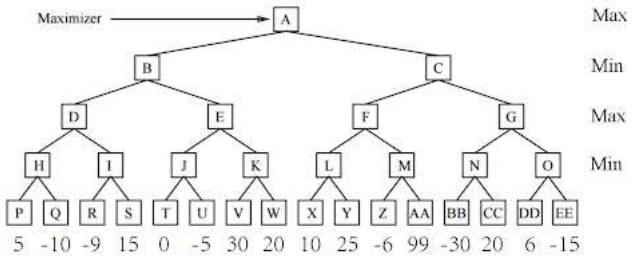
Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the A^* tree search algorithm with $f(n) = (2 + x) \cdot g(n) + (x^2) \cdot h(n)$
where $g(n)$ is the cost from the start state to state n and $h(n)$ is an admissible estimate of the cost from state n to the goal state.
For how many integer values of x is the algorithm guaranteed to be optimal?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the min-max tree in Figure, A to EE are numbers on the leaf, and the numbers for leaf are shown in the last row.



Which nodes get pruned if you use alpha-beta pruning when you explore successors of states going right-to-left?

A. K, V, W, AA, CC
B. E, J, K, T, V, W
C. O, DD, EE
D. L, P, T, X, Y

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

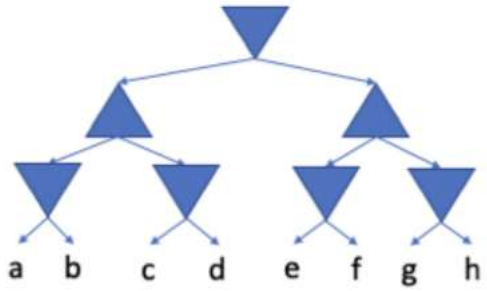
Consider the following two statements:
$\mathbf{S}_1$: The minimax procedure, without alpha-beta pruning, will always identify the best move for the player whose turn it is to move.
$\mathbf{S}_2$: Minimax with alpha-beta pruning will minimize the number of branches that are pruned if the moves under a node are ordered from best to worst.
Which of the following is correct?

A. S_1 is true, S_2 is false
B. S_1 is false, S_2 is true
C. Both are true
D. None of them is true

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the minimax game tree shown below. Decisions by MAX are represented as upward-pointing triangles; decisions by MIN are represented as downward-pointing triangles; small letters denote outcomes of the game:



The values of each of the outcomes, to the MAX player, are as shown in the following table:

	Outcome							
	a	b	c	d	e	f	g	h
Value of the MAX player:-	8	3	1	7	2	5	6	4

If the values of the two MAX nodes are x and y. What is $\mathrm{x}+\mathrm{y}$?

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

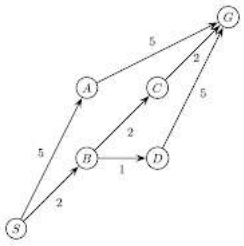
Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider an infinite search space $\mathbb{Z} \times \mathbb{Z}$. The start state is at $(0,0)$ and the goal state is at $\left(\mathrm{g}_{\mathrm{x}}, \mathrm{g}_{\mathrm{y}}\right)$. Given that the agent can move from a state (x, y) to either of $\{(\mathrm{x}+1, \mathrm{y}),(\mathrm{x}-1, \mathrm{y}),(\mathrm{x}, \mathrm{y}+1),(\mathrm{x}, \mathrm{y}-1)\}$ with a unit step cost, find the number of nodes explored using A^* tree search with Manhattan distance heuristic for $\left(\mathrm{g}_{\mathrm{x}}, \mathrm{g}_{\mathrm{y}}\right)=(6,7)$. Assume no duplicate detection.

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the following search problem:



Node	h_0	h_1	h_2
S	0	5	6
A	0	3	5
B	0	4	2
C	0	2	5
D	0	5	3
G	0	0	0

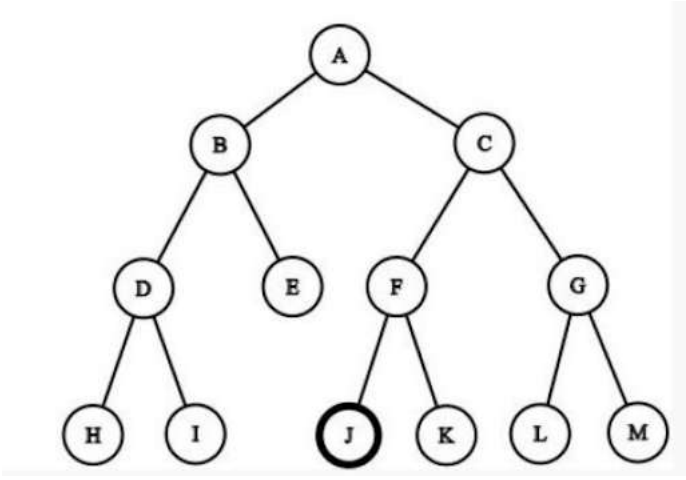
Which of the heuristics are admissible?

A. h_0
B. h_1
C. h_2
D. None of the above

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

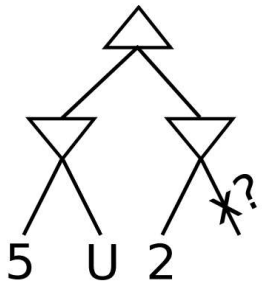
Consider the following graph in which we are searching from start state A to the goal state J. What is the number of nodes generated by Iterative Deepening Search?
Assume that nodes are explored in right-to-left order and all edge costs are 1.
Note: Exploration of the goal state J should be added to the count.



Your Answer: | Correct Answer: 19 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the game tree shown below. For which of the following values of U,will the indicated pruning take place?



A. $\mathrm{U} = 3$
B. $\mathrm{U} = 5$
C. $\mathrm{U} = 1$
D. $\mathrm{U} = -5$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following statements about search algorithms is/are true?

A. Breadth-First Search is optimal when all edge costs are equal and non-negative.
B. Depth-First Search is complete when performing graph search.
C. Iterative Deepening Search is optimal when all edge costs are equal and non-negative.
D. A^* Search is optimal when an admissible heuristic (in tree search) or a consistent heuristic (in graph search) is used.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the depth-limited search algorithm with depth limit f.
If the only goal state exists at depth d then which of the following statement(s) is/are correct?

A. If $\mathrm{f} > \mathrm{d}$ then it is complete but not optimal.
B. If $\mathrm{f} > \mathrm{d}$ then it is not complete and not optimal.
C. If $\mathrm{f} < \mathrm{d}$ then it is complete but not optimal.
D. If $\mathrm{f} < \mathrm{d}$ then it is not complete and not optimal.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

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